From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FULL TEST – I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 09)** contains 09 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (10 – 13) contains 4 Assertion-Reasoning (multiple choice questions) which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (14 – 19) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-B (1 – 03)** contains 3 Matrix Match Type (4 × 4 Matrix) questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+6 marks** for all correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

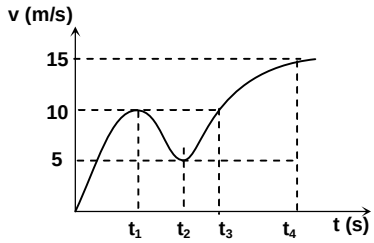
PART – I

SECTION – A

Single Correct Choice Type

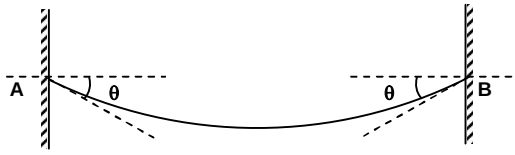
This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Velocity time graph of a particle undergoing rectilinear motion is plotted as shown in the figure. Average acceleration of particle is zero between time intervals.



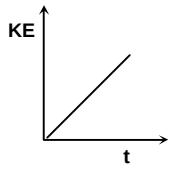
(A) 0 and t_1 (B) t_1 and t_2
(C) t_1 and t_3 (D) t_2 and t_4

2. A heavy string of mass m hangs between two fixed points A and B at the same level. The tangents to the string at A and B are at an angle θ with the horizontal as shown in the figure. The tension in the string at lowest point is



(A) $\frac{mg}{2\sin\theta}$ (B) $\frac{mg}{2\cos\theta}$
(C) $\frac{mg}{2\tan\theta}$ (D) $\frac{mg}{2\cot\theta}$

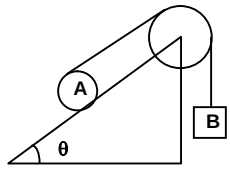
3. The kinetic energy of a body moving along a straight line varies with time as shown in the figure. The force acting on the body is



(A) zero
(B) constant
(C) directly proportional to velocity
(D) inversely proportional to velocity.

Space for rough work

4. In the figure shown, a string is wound over a cylinder A. The other end of the string is attached to block B through a pulley. When the system is released the cylinder rolls down without slipping. The ratio of magnitude of vertical component of displacement of A and B in any time interval t is



(A) $\sin\theta : 1$ (B) $\sin\theta : 2$

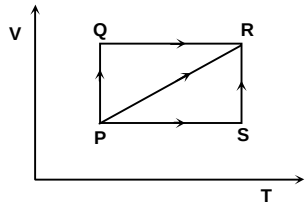
(C) $\cos\theta : 1$ (D) $\cos\theta : 2$

5. Six stars of equal mass are moving about the centre of mass of the system such that they are always on the vertices of a regular hexagon of side length a. Their common time period will be

(A) $4\pi\sqrt{\frac{a^3}{Gm}}$ (B) $2\pi\sqrt{\frac{4\sqrt{3}a^3}{Gm\left(5\sqrt{3}+4\right)}}$

(C) $4\pi\sqrt{\frac{3a^3}{Gm}}$ (D) None of these

6. One mole of a monoatomic gas is taken from P to R, via three paths PQR, PR and PSR. If work done by the gas in PQR is W_1, in PR work done is W_2 and in PSR work done is W_3, then



(A) $W_2 > W_3 > W_1$ (B) $W_1 > W_2 > W_3$

(C) $W_2 > W_1 > W_3$ (D) $W_3 > W_2 > W_1$

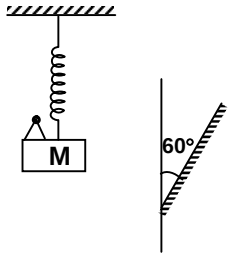
7. A block of wood of relative density 0.5 is placed 10 m inside a vessel containing water. The vessel is accelerated upward with an acceleration of 10 m/s^2. If the block is released at some instant, then the time taken by block to reach the surface of water from that instant is (g = 10 m/s^2)

(A) 0.5 sec (B) 1 sec

(C) 2 sec (D) none of these

Space for rough work

8. An insect of negligible mass is sitting on a block of mass M, tied with a spring of force constant k. The block performs simple harmonic motion with amplitude A infront of a plane mirror placed as shown in the figure. The maximum speed of insect relative to its image will be



(A) $A\sqrt{\frac{k}{m}}$ (B) $\frac{A\sqrt{3}}{2}\sqrt{\frac{k}{m}}$

(C) $A\sqrt{3}\sqrt{\frac{k}{m}}$ (D) $A\sqrt{\frac{m}{k}}$

9. Two concentric conducting spheres of radii R and 3R carrying charges Q and 2Q respectively. If the charge on inner sphere is doubled. The potential difference between inner and outer spheres will

(A) becomes two times (B) becomes four times

(C) be halved (D) remains same

Assertion - Reasoning Type

This section contains 4 questions numbered 10 to 13. Each question contains **STATEMENT-1 (Assertion)** and **STATEMENT-2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

10. STATEMENT-1: The time period of a pendulum of infinite length whose bob hangs near the surface of the earth will be infinite.

and

STATEMENT-2: The time period of a pendulum of length L near the surface of the earth is $2\pi\sqrt{\frac{L}{g}}$, if L is reasonably small .

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.

(C) Statement -1 is True, Statement-2 is False.

(D) Statement -1 is False, Statement-2 is True.

Space for rough work

11. STATEMENT-1: Action and reaction are always equal and opposite, they act on different bodies and they act along the same line (They are collinear)
and

STATEMENT-2: The summation of torques of internal forces in any rigid body is always zero.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

12. STATEMENT-1: An electromagnetic induction the Lenz law tells us only about the direction of induced emf while Faraday's law give us only the magnitude of emf.
and

STATEMENT-2: The direction of induced emf is such that it tends to oppose the cause of its production.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

13. STATEMENT-1: In X-rays the energy of K_α photon is lower than the energy of a K_β photon.
and

STATEMENT-2: The number of K_α photons is much larger than K_β photons.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

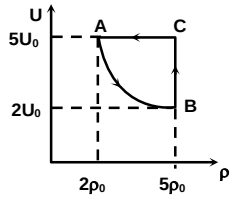
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Comprehension Type

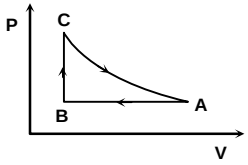
This section contains 2 groups of questions. Each group has **3 multiple choice question** based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

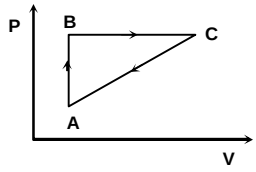
Paragraph for Question Nos. 14 to 16

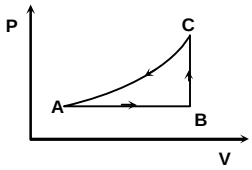
Figure shows the variation of the internal energy U with the density ρ of one mole of ideal monoatomic gas for a thermodynamics cycle ABCA. Here process AB is a part of rectangular hyperbola



14. The P-V diagram of above process

(A) 

(B) 

(C) 

(D) None of these

15. The total amount of heat absorbed by the system for cyclic process is

(A) $\left(\frac{10}{3}\ln 2.5 - 2\right)U_0$ (B) $\left(\frac{10}{5}\ln 0.4 - 2\right)U_0$

(C) $50U_0$ (D) None of these

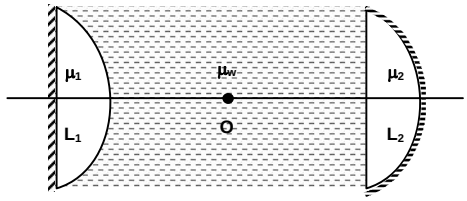
16. The work done in process AB is

(A) $-U_0$ (B) $-2U_0$

(C) $-5U_0$ (D) none of these.

Space for rough work

Paragraph for Question Nos. 17 to 19



A cylindrical tube filled with water ($\mu_w = 4/3$) is closed at its both ends by two silvered plano convex lenses as shown in the figure. Refractive index of lenses L_1 and L_2 are 2.0 and 1.5 while their radii of curvature are 5 cm and 9 cm respectively. A point object is placed some where at a point O on the axis of cylindrical tube. It is found that the object and image coincide each other.

17. The position of object w.r.t. lens L_1 is
(A) 8 cm (B) 10 cm
(C) 12 cm (D) 14 cm

18. The position of object w.r.t. lens L_2 is
(A) 8 cm (B) 10 cm
(C) 12 cm (D) 14 cm

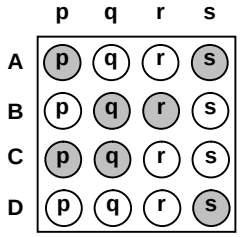
19. The length of the cylindrical tube is
(A) 16 cm (B) 18 cm
(C) 20 cm (D) 22 cm

Space for rough work

SECTION - B

Matrix – Match Type

This section contains 3 questions. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.



If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4×4 matrix should be as follows:

1. Three concentric spherical metallic shells A, B, and C of radii a, b, and c (a<b<c) have charge densities of σ, $-\sigma$ and σ respectively, then match the following:

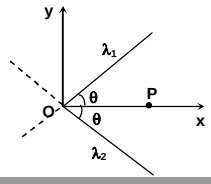
Column I		Column II	
(A)	The potential at A	(p)	$\frac{\sigma}{\varepsilon_0}\left(\frac{a^2}{c}-\frac{b^2}{c}+c\right)$
(B)	The potential at B	(q)	$\frac{\sigma}{\varepsilon_0}\left(\frac{a^2}{c}-b+c\right)$
(C)	The potential at C	(r)	$\frac{\sigma}{\varepsilon_0}$
(D)	The electric field at surface of A	(s)	$\frac{\sigma}{\varepsilon_0}(a-b-c)$

Space for rough work

2. For an atom having atomic number z.
Match the following:

Column I		Column II	
(A)	Radius of orbit	(p)	is proportional to z
(B)	Current associated due to orbital motion of electron	(q)	is inversely proportional to z.
(C)	Magnetic field at the centre due to orbital motion of electron	(r)	is proportional to z^2.
(D)	Velocity of an electron	(s)	is proportional to z^3

3. Two infinitely long line charges λ_1 and λ_2 are placed symmetric to x-axis as shown in the figure. Match the following for direction of electric field at point P on the x-axis.



Column I		Column II	
(A)	Both λ_1 and λ_2 are positive	(p)	positive x-axis
(B)	Both λ_1 and λ_2 are negative	(q)	positive y-axis
(C)	λ_1 is positive but λ_2 is negative	(r)	negative x-axis
(D)	λ_1 is negative but λ_2 is positive	(s)	negative y-axis

Space for rough work

Chemistry

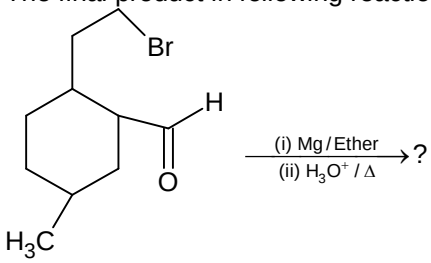
PART – II

SECTION – A

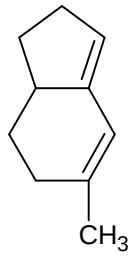
Straight Objective Type

This section contains 9 multiple choice questions numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

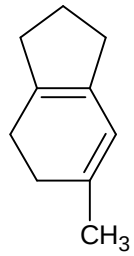
1. The final product in following reaction is



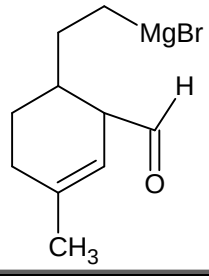
(A)



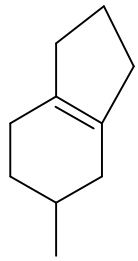
(B)



(C)

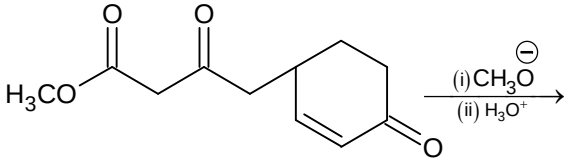


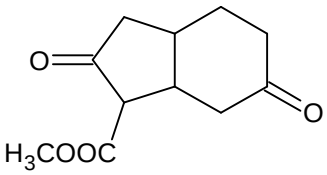
(D)

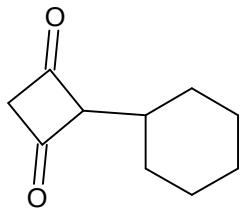


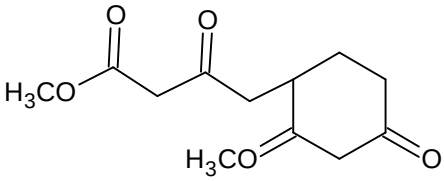
Rough Work

2. Find out major product of following reaction:



(A) 

(B) 

(C) 

(D) None of these

3. At a given temp, total vapour pressure in Torr of a mixture of volatile components A and B is given by $P = 120 - 75 \times B$.
Hence vapour pressures of pure A and B respectively (in Torr) are
(A) 120,75 (B) 120,195
(C) 120,45 (D) 75, 45

4. Equivalent conductance of saturated solution of $BaSO_4$ is 400 $ohm^{-1}cm^2eqv^{-1}$ and specific conductance is 8×10^{-5} $ohm^{-1}cm^{-1}$. Hence Ksp of $BaSO_4$ is
(A) 4×10^{-8} m^2 (B) 1×10^{-8} m^2
(C) 2×10^{-4} m^2 (D) 1×10^{-4} m^2

5. Number of natural life times (Tav) required for a first order reaction to achieve 99.9 level of completion is:
(A) 2.3 (B) 6.9
(C) 9.2 (D) 0.105

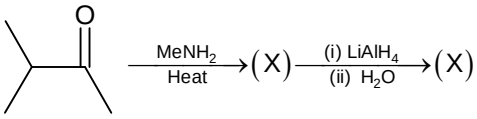
6. In $[B_4O_5(OH)_4]^{2-}$, the number of boron atoms having an octet of electron is:
(A) 0 (B) 1
(C) 2 (D) 4

Rough Work

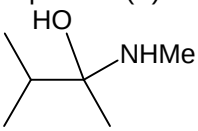
7. Maximum percentage of available chlorine on the basis of $Ca(OCl)Cl.H_2O$ formula is
(A) 35 (B) 40
(C) 45 (D) 49

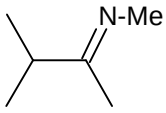
8. In which of the following pairs of molecules have bond order three and are isoelectronic?
(A) NO^+ , CO^+ (B) CN^- , CO
(C) CN^- , O_2^+ (D) CO, O_2^+

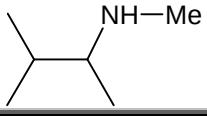
9. Consider the following sequence of reactions

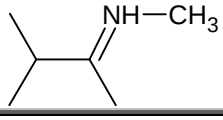


The final product (X) is

(A) 

(B) 

(C) 

(D) 

Rough Work

Reasoning Type

This section contains 4 questions numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

10. STATEMENT-1: The equilibrium constant for a reaction having standard heat reaction ΔH^o increases with temperature.
and
STATEMENT-2: The temperature dependence of equilibrium constant is related to ΔS^o and ΔH^o for the reaction.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

11. STATEMENT-1: Isomerisation also results when $NaNH_2$ is used in dehydrbromination of 2, 2 dibromobutane forming 2-butyne as well as 1-butyne.
and
STATEMENT-2: All possible triple bond isomers are formed but $NaNH_2$ is such a strong base that it deprotonates the terminal alkyne removing it from the equilibrium.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Rough Work

12. STATEMENT-1: Green solution of potassium magnate (VI) K_2MnO_4 turns purple and brown solid is precipitated when CO_2 bubbled into it.
and
STATEMENT-2: K_2MnO_4 changes to K_3MnO_8 by CO_2.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

13. STATEMENT-1: Melting points of alkanes with even number of carbon atoms are higher than those of alkanes with odd number of carbon atoms.
and
STATEMENT-2: Alkanes with even number of carbon atom fit together better in crystal lattice.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

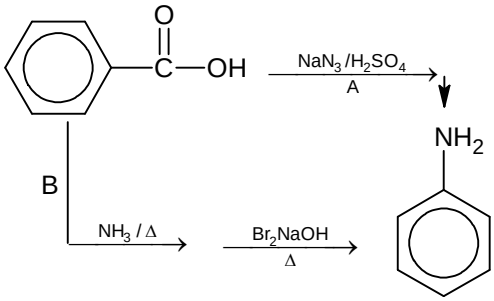
Rough Work

Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

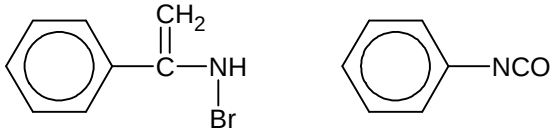
Paragraph for Question Nos. 14 to 16

Benzoic acid can be converted into aniline by different reagents



Following questions are based on the above reactions

14. Consider the following facts about reaction (B)
Fact (P): It is intramolecular migration reaction
Fact (Q): It is degradation reaction.
Fact (R): It involves following intermediates.



Select correct facts

(A) P, Q
(B) Q and R
(C) P, R
(D) P, Q, R

15. Escaping gases in reaction (A) is/are:

(A) NH_3, CO_2
(B) N_2, NH_3
(C) N_2, CO_2
(D) CO_2 only

16. These reactions are called

	A	B
(A)	Hoffman	Schmidt
(B)	Schmidt	Hoffmann
(C)	HVZ	Hoffman
(D)	Hoffman	HVZ

Rough Work

Paragraph for Question Nos. 17 to 19

The following reaction was studied at 25°C in benzene solution containing 0.1 M pyridine.

$$\underset{(A)}{CH_3OH} + \underset{(B)}{(C_6H_5)_3CCl} \longrightarrow \underset{(C)}{(C_6H_5)_3COCH_3} + HCl$$

The following sets of data were observed.

Set	Initial concentration		Time	Final concentration (C)
	$[A]_o$	$[B]_o$		
I	0.10 M	0.05 M	25 min	0.0033 m
II	0.10 M	0.10 M	15 min	0.0039 m
III	0.20 M	0.10 M	7.5 min	0.0077 m

17. Rates $\frac{d[C]}{dt}$ in sets (I), II and III are respectively (in M min^{-1})

	I	II	III
(A)	1.30×10^{-4}	2.6×10^{-4}	1.02×10^{-3}
(B)	0.033	0.0039	0.0077
(C)	0.02×10^{-4}	0.04×10^{-4}	0.0077
(D)	None of these		

18. Rate law of above experiment is

(A) $K[A][B]$ (B) $K[A]^2[B]$

(C) $K[A][B]^2$ (D) $K[A]^2[B]^4$

19. Rate constant of the above experiment is (in $L^2m^{-2}min^{-1}$)

(A) 1.3×10^{-1} (B) 2.6×10^{-2}

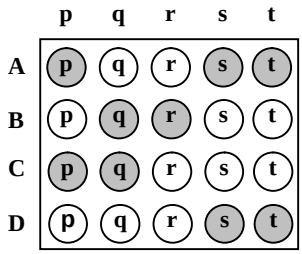
(C) 2.6×10^{-1} (D) 1.3×10^{-2}

Rough Work

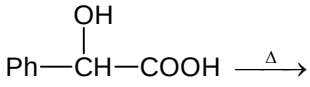
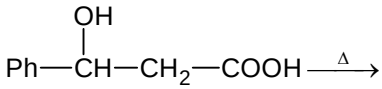
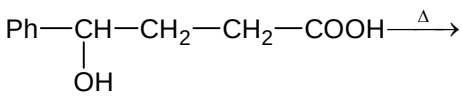
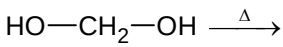
SECTION-B
(Matrix Type)

This section contains 3 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

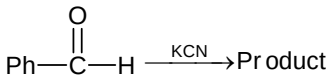
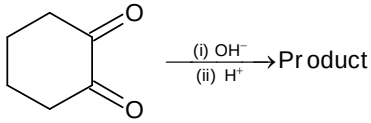
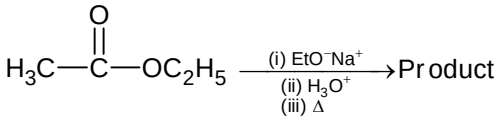
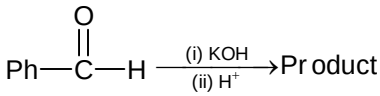
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following

(A)		(p)	Cyclic
(B)		(q)	Exibit geometrical isomerism.
(C)		(r)	Can be optically active
(D)		(s)	Lactonisation

2. Match the following

	Column – I		Column – II
(A)		(p)	Final product gives positive Tollen's test.
(B)		(q)	Final product gives test with 2,4 – DNP
(C)		(r)	Final product reacts with Na and liberates CO_2 gas
(D)		(s)	Final product reacts with Na and liberates H_2 gas.

3. Match the radicals of column – I with their group reagents in column II

	Column – I		Column – II
(A)	Pb^{2+}	(p)	H_2S
(B)	Co^{2+}	(q)	$(NH_4Cl + NH_4OH)$
(C)	Fe^{3+}	(r)	HCl
(D)	Ba^{2+}	(s)	$(NH_4)_2CO_3$

Rough Work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **9 multiple choice questions** numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Consider a 9 sided regular polygon. If 5 vertices are selected at random then the minimum pairs of parallel lines formed from those 5 vertices is
   (A) 0 (B) 1
   (C) 2 (D) none of these

2. Consider a curve $|x| + |y| = 1$ such that lines $y = mx$, $y = nx$ make points of intersection in the same quadrant. Let A, B be two such points lying in the same quadrant such that OA, AB, OB are in A.P. then the maximum angle θ is (0 is origin)
   (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
   (C) $\frac{\pi}{4}$ (D) none of these

3. Let, $a_1, a_2, a_3, \ldots\ldots a_{23}$ be 23 integers (not necessarily distinct). Then, the prime number which always divides the sum of elements of atleast one subset of these numbers can be
   (A) 23 (B) 29
   (C) 31 (D) 37

4. Given that $|z_1|, |z_2|, |z_3|, |z_4|$ are non zero positive real numbers where z_1, z_2, z_3, z_4 are complex numbers then $\frac{1}{|z_1|}+\frac{1}{|z_2|}+\frac{4}{|z_3|}+\frac{16}{|z_4|}=\frac{k}{|z_1|+|z_2|+|z_3|+|z_4|}$ where k is
   (A) 8 (B) 16
   (C) 64 (D) none of these

Space for rough work

5. Let, $C_1 : r = ae^{\theta}$ and $C_2 : re^{\theta} = b$ be two curves then the angle between the curves is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

6. Given the equation $f''(x) = 0$; f, f′, f″ are continuous in $a \le x \le b$ and curve $y = f(x)$ crosses the x–axis at least in 3 distinct points in [a, b] then which of the following represents the minimum number of roots in (a, b)

(A) 2 (B) 3

(C) 4 (D) none of these

7. The possible solution of the differential equation $y(y^2 - 2x^2)dx + x(2y^2 - x^2)dy = 0$ is

(A) $x^2y^2(x^2 + y^2) = c$ (B) $x^2(y^2 - x^2) = c$

(C) $x^2y^2(y^2 - x^2) = c$ (D) none of these

8. A circle with radius 'a' and centre on y-axis slides along y-axis and a variable line through (a, 0) cuts the circle at points P and Q, then the point of intersection of tangents to the circle at P and Q will lie in the region defined by

(A) $y^2 \ge 4a(x - a)$ (B) $y^2 \le 4ax$

(C) $x^2 + y^2 \le 4a^2$ (D) $x^2 - y^2 \ge a^2$

9. If f: [0, 1] $\to$ (0, ∞) such that $\int_0^1 f(x)dx = 1$, $\int_0^1 xf(x)dx = \alpha$ and $\int_0^1 x^2 f(x)dx = \alpha^2$, then

(A) f(x) = 5 (B) f(2) = 8

(C) f(x) is not possible (D) f(x) = (2x – 1)

Space for rough work

Reasoning Type

This section contains **4 questions** numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

10. STATEMENT 1: $2p^3 - (2x + 4 \sin x - \cos x)p^2 - (x \cos x - 4x \sin x + \sin 2x)p + x \sin 2x = 0$ is a differential equation with a possible solution as $(2y - x^2 - c)(y + 2 \cos x - c)$ $(2y + \sin x - c) = 0$; $\left[p = \frac{dy}{dx}\right]$

STATEMENT 2: Degree of differential equation is the power of the highest order derivative existing in the differential equation

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

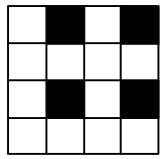
11. STATEMENT 1: Consider a polynomial such that $p(\lambda) = \frac{\lambda}{\lambda + 1}$, $\lambda \in I^+ \cup \{0\}$, $\lambda \le n$ then

$p(n+1) = \frac{n}{n+2}$

STATEMENT 2: Let, $Q(x) = (x + 1)\, p(x) - x$ then $Q(x)$ vanishes for $x = 0, 1, 2, \ldots, n$ hence,
$(x + 1)\, p(x) - x = \alpha x(x - 1)(x - 2) \ldots (x - n)$

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

12. STATEMENT 1: A rectangular floor is covered by two kinds of tiles 2 × 2 and 1 × 4. One tile is smashed and there is a tile of other kind available then, the floor cannot be covered by rearranging the tiles



STATEMENT 2: If we colour the floor as shown 4 × 1 tile covers 0 or 2 black squares. A 2 × 2 tile always covers one black square so, it is impossible to replace one type of tile for the other

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

Space for rough work

13. STATEMENT 1: Consider an ellipse $ax^2 + 2hxy + by^2 = 1$ then its eccentricity (e) is

$$e^2 = \frac{\sqrt{(a-b)^2+4h^2}}{2(ab-h^2)}\left[(a+b)-\sqrt{(a-b)^2+4h^2}\right]$$

STATEMENT 2: Any ellipse can be transformed to the principal axis as $\frac{x'^2}{\alpha^2}+\frac{y'^2}{\beta^2}=1$ then associated eccentricity is $e^2 = \frac{\alpha^2+\beta^2}{\alpha^2}$

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:
Consider a function $f : R \to R$ such that $(f(x) + f(z))(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$ where $x, y, z, t \in R$

14. For the function, $f : R \to R$, f(x) can be
(A) even (B) odd
(C) neither even nor odd (D) none of these

15. If $f : R \to R$ is the given function then in the interval $(0, \infty)$ f(x) is
(A) non decreasing (B) non increasing
(C) decreasing (D) none of these

16. The number of functions $f : R \to R$ which satisfy the given functional equation is
(A) 1 (B) 2
(C) 3 (D) none of these

Space for rough work

Paragraph for Question Nos. 17 to 19

Read the following write up carefully and answer the following questions:

Consider a parabola, $y^2 = 4ax$ then $y + tx = at^3 + 2at$ represents a normal at a point $(at^2, 2at)$. It can also be represented as $y - mx + am^3 + 2am = 0$ in its slope form.

Considering the above equation and assuming it passes through a point P(h, k) we can say that $k - mh + am^3 + 2am = 0 \Rightarrow am^3 + m(2a - h) + k = 0$ from P(h, k) maximum 3 normals can be made to the parabola $y^2 = 4ax$

17. Consider a point $P(x_1, 0)$ lying inside the parabola $y^2 = 4ax$ then the interval of x_1 such that only one normal passes through P is

(A) (0, 8a) (B) (0, 2a)

(C) 0, 4a) (D) none of these

18. The number of points for the parabola $y^2 = 4ax$ from which 3 coincident normals can be drawn is

(A) 1 (B) 2

(C) 4 (D) none of these

19. The area of the region inside the parabola $y^2 = 4ax$ from which only one normal can be drawn is

(A) $\frac{352\sqrt{2}a^2}{15}$ (B) $32\sqrt{2}a^2$

(C) $\frac{48\sqrt{2}a^2}{5}$ (D) none of these

SECTION – B
(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	If 2 is the root of the equation $\|A - xI\| = 0$, (where A is a non singular matrix), $\frac{\|A\|}{2}$ a root of $\|B - xI\| = 0$, then B can be	(p)	e
(B)	If $e^{i\alpha}$ is the root of $\|A - yI\| = 0$ then a root of $\|\bar{A}' - xI\| = 0$ is (where A is a non singular matrix)	(q)	adj(A)
(C)	Let A_{ij} be a 2×2 non singular matrix where $i, j \in N$ and $B = \begin{bmatrix} \|A_{11}\| & \|A_{12}\| & \ldots\ldots\ldots & \|A_{1n}\| \\ 0 & \|A_{22}\| & \ldots\ldots\ldots & \|A_{2n}\| \\ 0 & 0 & \|A_{33}\| \ldots & \|A_{3n}\| \\ - & - & - & - \\ - & - & - & - \\ 0 & - & - & \|A_{nn}\| \end{bmatrix}$ then $\|B - \lambda I\| = 0$ has root as	(r)	$\cos\alpha - i\sin\alpha$
(D)	Consider a matrix such that $\bar{A}' = A$ then the equation $\|A - xI\| = 0$ can have root as (where A is a non singular matrix)	(s)	$\|A_{11}\|$

Space for rough work

2. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	If $32 \sin^4\theta \cos^2\theta = \cos 6\theta - 2\cos 4\theta - \cos 2\theta + \lambda$ then λ is	(p)	$\frac{1}{8}$
(B)	Consider a tetrahedron with just one edge of length greater than 1 then its maximum volume is	(q)	$\frac{3}{2}$
(C)	Consider an acute angled triangle ABC such that $x_n = 2^{n-3}(\cos^n A + \cos^n B + \cos^n C) + \cos A \cos B \cos C$ where $n \in N$ then $x_1 + x_2 + x_3$ is atleast	(r)	$\frac{8}{9}$
(D)	Consider a ΔABC such that $\left(\cot\frac{A}{2}\right)^2 + \left(2\cot\frac{B}{2}\right)^2 + \left(3\cot\frac{C}{2}\right)^2 = \left(\frac{6}{7}\frac{S}{r}\right)^2$ where S and r are semi perimeter and inradius respectively, then $\frac{b}{c}$ is	(s)	2

3. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	Consider an equation $8x^4 - 16x^3 + 16x^2 - 8x + a = 0$ then the sum of all the non real roots of the equation can be ($a \in R$)	(p)	9
(B)	If $x, y \in R^+$ such that $x + y = 2$ then the greatest value of $2x^3y^3(3x^3 + 3y^3)$ is	(q)	18
(C)	The number of all 7–digit numbers formed by using only the digits 5 and 7 & divisible by both 5 and 7 are	(r)	12
(D)	$X = \{8, 9, 10, \ldots..\}$; $f : X \to X$ and $f(x + y) = f(xy)\ \forall\ x \geq 4$, $y \geq 4$; $f(8) = 9$ then $2f(9)$	(s)	2

Space for rough work